

677 Final Review

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1 Representation Theory

Claim: Let V be over F with $\text{char } F \nmid |G|$. Let $U \leq V$ be a subrep. Then there exists a subrep $U' \leq V$ such that $U \oplus U' = V$.

Proof: First, let $\varphi : V \rightarrow W$ be a homrep. I claim that $\ker \varphi \leq V, \text{im } \varphi \leq W$ are subreps. $\ker \varphi \leq V$ is a subspace. Let $v \in \ker \varphi$ and $g \in G$. Then $\varphi(gv) = g\varphi(v) = g0 = 0$, so $gv \in \ker \varphi$. Let $w \in \text{im } \varphi$, so $w = \varphi(v)$ for some $v \in V$. Then $gw = g\varphi(v) = \varphi(gv)$.

There exists a contraction $\pi : V \rightarrow U$. Define $\pi' : V \rightarrow U$ by $\pi'(v) = \frac{1}{|G|} \sum_{g \in G} g^{-1} \pi(gv)$. For $u \in U$, we have $\pi'(u) = \frac{1}{|G|} \sum_{g \in G} g^{-1} gu = u$, so U is fixed. Further,

$$h\pi'(v) = \frac{1}{|G|} \sum_{g \in G} hg^{-1} \pi(gv) = \frac{1}{|G|} \sum_{g \in G} g^{-1} \pi(ghv) = \pi'(hv),$$

so π' is G -linear. By Linear Algebra, $\ker \pi' \oplus \text{im } \pi' = V$. Note $\text{im } \pi' = U$ since U is fixed. We take $U' = \ker \pi'$, which is a subrep.

Claim: Let V/F be a rep. Then V decomposes as a direct sum of irreducible subreps. A rep U is said to be irreducible if U has no nontrivial subreps. This shows that irreducible and indecomposable subreps are the same.

Proof: I induct on $\dim V$. If $\dim V = 0$, we're done. If V is irreducible, we're done. Else, let U be a nontrivial subrep. Let U' be a complementary subrep. Write $V = U \oplus U'$. By the IH, U, U' are direct sums of irreps, so V is a direct sum of irreps.

Claim: Let $V, W/\mathbb{C}$ be irreps of G . Then

$$\text{Hom}_G(V, W) = \begin{cases} \mathbb{C} & \text{if } V \cong W, \\ 0 & \text{if } V \not\cong W. \end{cases}$$

Proof: Let $\varphi : V \rightarrow W$ be G -linear. $\ker \varphi \leq V$. If $\ker \varphi = V$, we have $\varphi = 0$. Else, $\ker \varphi = 0$, so φ is injective. $\text{im } \varphi \leq W$. If $\text{im } \varphi = 0$, then $\varphi = 0$. Else, φ

is surjective. Then either $\varphi = 0$ or φ is an isomorphism.

Hence, take $V = W$. Suppose $\varphi : V \rightarrow V$ is a G -linear map. Since $\mathbb{C}$ is algebraically closed, φ has an eigenvalue λ . That is, $\ker(\varphi - \lambda 1) \neq 0$, so $\ker(\varphi - \lambda 1) = V$, so $\varphi = \lambda 1$. This choice of λ is unique for each φ . Denote this choice by Φ . It is easy to see that $\Phi : \text{Hom}_G(V, V) \rightarrow \mathbb{C}$ is a field isomorphism.

Claim: A decomposition into irreps is unique up to multiplicities over $\mathbb{C}$.

Proof: Write as a sum of irreducibles $V = \bigoplus_{i \in [r]} V_i^{a_i} = \bigoplus_{i \in [r]} V_i^{b_i}$, where $a_i, b_i \in \mathbb{Z}_{\geq 0}$. We have $\text{Hom}_G(V, V_j) = \bigoplus_{i \in [r]} \text{Hom}_G(V_i, V_j)^{\oplus a_i} = \mathbb{C}^{\oplus a_j} = \mathbb{C}^{\oplus b_j}$. Since $a_i, b_i < \infty$, we have $a_i = b_i$ for all $i \in [r]$.

Define $\chi : G \rightarrow \mathbb{C}$ by $\chi(g) = \text{tr } g$. Note that $\text{tr } h^{-1}gh = \text{tr } g$, so tr is a class function. Given two class functions $f_1, f_2 : G \rightarrow \mathbb{C}$, define $\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g)$. Then $\langle f_1 + f_1', f_2 \rangle = \langle f_1, f_2 \rangle + \langle f_1', f_2 \rangle$ and $\langle \lambda f_1, f_2 \rangle = \lambda \langle f_1, f_2 \rangle$.

Given V a G -rep, denote by V^G the set of points in V fixed by G . Note $0 \in V^G$, if $x, y \in V^G$, then $g(x + y) = gx + gy = x + y$, so $x + y \in V^G$, and if $\lambda \in \mathbb{C}$, then $g(\lambda x) = \lambda gx = \lambda x$, so $\lambda x \in V^G$. Finally, if $x \in V^G$, then $gx = x \in V^G$, so $V^G \leq V$.

Define $\pi : V \rightarrow V$ by $\pi(v) = \frac{1}{|G|} \sum_{g \in G} gv$. Note that if $v \in V^G$, then $\pi v = v$, so V^G is fixed by π . Further,

$$h\pi(v) = \frac{1}{|G|} \sum_{g \in G} hg(v) = \frac{1}{|G|} \sum_{g \in G} gv = \pi(v),$$

so π is a retract of V to V^G . Note that $h\pi(v) = \pi(hv) = \pi(v)$, so π is G -linear.

Let $\{e_i\}_{i \in [m]}$ be a basis of V^G and extend to a basis $\{e_i\}_{i \in [n]}$ of V . Then the matrix of π with respect to $\{e_i\}$ has m ones on the diagonal, so $\text{tr } \pi = \dim V^G$. We compute

$$\text{tr } \pi = \frac{1}{|G|} \sum_{g \in G} \text{tr } g = \frac{1}{|G|} \sum_{g \in G} \chi_V(g),$$

so $\sum_{g \in G} \chi_V(g) = |G| \dim V^G$.

Consider the vector space $\text{Hom}(V, W)$, where V, W are G -reps. Note that the elements of $\text{Hom}(V, W)$ need not be G -linear. G acts on $\text{Hom}(V, W)$ by $g\varphi = g\varphi g^{-1}$. Note that

$$gh\varphi = gh\varphi(gh)^{-1} = gh\varphi h^{-1}g^{-1} = g(h\varphi)$$

(the map $\varphi \mapsto g^{-1}\varphi g$ is not well-defined). If φ is fixed by all g , then $g\varphi g^{-1} = \varphi$, ie φ is G -linear.

Claim: $\text{Hom}(V, W) \cong V^* \otimes W$ as representations.

Proof: Define $\Phi : V^* \otimes W \rightarrow \text{Hom}(V, W)$ by $\Phi(v^* \otimes w)(x) = v^*(x)w$. Note that

$$\begin{aligned} g\Phi(v^* \otimes w)(x) &= g(v^*(g^{-1}x)w) = v^*(g^{-1}x)gw \\ \Phi(gv^* \otimes gw)(x) &= gv^*(x)gw = v^*(g^{-1}x)gw, \end{aligned}$$

so Φ is G -linear. Note the action on V^* is given by $gv^* = v^*g^{-1}$ so that $ghv^* = v^*h^{-1}g^{-1} = g(hv^*)$.

Pick bases $\{v_i^*\}_{i \in [n]}, \{w_j\}_{j \in [m]}$, so $\{v_i^* \otimes w_j\}_{(i,j) \in [n] \times [m]}$ is a basis of $V^* \otimes W$. I claim that the $\Phi(v_i^* \otimes w_j)$ form a basis of $\text{Hom}(V, W)$. Note that $\dim \text{Hom}(V, W) = \dim V \cdot \dim W = \dim V^* \cdot \dim W = \dim(V^* \otimes W)$. Further if for all $x \in V$ we have

$$\begin{aligned} \sum_{ij} \lambda_{ij} \Phi(v_i^* \otimes w_j)(x) &= 0 \rightarrow \sum_{ij} \lambda_{ij} v_i^*(x) w_j = 0 \\ &\rightarrow \sum_{i, j_0} \lambda_{ij_0} v_i^*(x) = 0 \rightarrow \lambda_{i_0 j_0} = 0 \end{aligned}$$

for all i_0, j_0 . Hence, Φ is an isomorphism of representations.

It follows that

$$\begin{aligned} \dim \text{Hom}_G(V, W) &= \dim \text{Hom}(V, W)^G = \frac{1}{|G|} \sum_{g \in G} \chi_{\text{Hom}(V, W)}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{V^* \otimes W}(g) = \frac{1}{|G|} \sum_{g \in G} \chi_{V^*}(g) \chi_W(g) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \langle \chi_V, \chi_W \rangle. \end{aligned}$$

In particular, if V, W are irreps, the claim from before follows.

Claim: There are finitely many irreps and their characters are linearly independent.

Proof: Let $\{V_i\}_{i \in [r]}$ denote a finite collection of distinct (up to isomorphism) irreps such that $\sum_{i \in [r]} \lambda_i \chi_{V_i} = 0$. Then

$$\left\langle \chi_{V_j}, \sum_{i \in [r]} \lambda_i \chi_{V_i} \right\rangle = \sum_{i \in [r]} \lambda_i \langle \chi_{V_j}, \chi_{V_i} \rangle = \lambda_j = 0,$$

so the χ_{V_i} are independent. In particular, the χ_{V_i} are distinct. Denote by $\{V_\alpha\}$ the set of all irreps. We have an embedding $\{V_\alpha\} \hookrightarrow \{\text{class functions on } G\}$.

Hence, there are at most $|\mathcal{C}|$ irreps, where $\mathcal{C}$ denotes the set of conjugacy classes of G .

Claim: $V \cong V'$ iff $\chi_V = \chi_{V'}$.

Proof: Write $V = \bigoplus_{i \in [r]} V_i^{\oplus a_i}$, $V' = \bigoplus_{i \in [r]} V_i^{\oplus a'_i}$. Then

$$\sum_{i \in [r]} a_i \chi_{V_i} = \sum_{i \in [r]} a'_i \chi_{V_i} \rightarrow a_i = a'_i \forall i \in [r],$$

so V, V' have the same decomposition into irreps, ie $V \cong V'$.

From now on, many details will be omitted. We will prove that the number of irreps equals $|\mathcal{C}|$ and compute the irreps of S_3 .

Let V be a G -rep and $f : G \rightarrow \mathbb{C}$ a function. Define $\varphi_{f,V} = \sum_{g \in G} f(g)g$. I claim that if f is a class function, then $\varphi_{f,V}$ is G -linear. Conversely, if $\varphi_{f,V}$ is G -linear for all V , then f is a class function.

Proof: We have

$$h\varphi_{f,V} = \sum_{g \in G} f(g)hg = \sum_{g \in G} f(g)gh = \varphi_{f,V}h.$$

Conversely, take $V = \mathbb{C}[G]$. Then $h\varphi_{f,\mathbb{C}[G]} = \varphi_{f,\mathbb{C}[G]}h$ implies $f(g) = f(hgh^{-1})$ for all $g, h \in G$.

Claim: Let $\alpha : G \rightarrow \mathbb{C}$ be a class function such that $\langle \alpha, \chi_W \rangle = 0$ for all irreps W . Then $\alpha = 0$.

Proof: Consider $\varphi_{\bar{\alpha},W} = \sum_{g \in G} \overline{\alpha(g)}g$, so

$$\text{tr } \varphi_{\bar{\alpha},W} = \sum_{g \in G} \overline{\alpha(g)} \text{tr } g = \sum_{g \in G} \overline{\alpha(g)} \chi_W(g) = \langle \alpha, W \rangle = 0,$$

so $\varphi_{\bar{\alpha},W}, \varphi_{\alpha,W}$ are identically zero. Since $\varphi_{\alpha,W} = \varphi_{\alpha,V}|_W$, we have $\varphi_{\alpha,V} = 0$ for all V . Take $V = \mathbb{C}[G]$, so $\alpha = 0$.

Induced Representations. Let V be a H -rep with $H \leq G$, so V is a left- $\mathbb{C}[H]$ module. By extension of scalars, $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} V$ is a left- $\mathbb{C}[G]$ module, ie a G -rep. Fix coset representatives $\{g_i\}_{i \in [n]}$ of G/H . I claim that $\{g_i\}$ is a basis of $\mathbb{C}[G]$ over $\mathbb{C}[H]$.

Claim: Left coset reps are the same as right coset reps.

Proof: Let $\{g_i\}$ be left coset reps. Suppose $Hg_i = Hg_j$. Then $g_i g_j^{-1} \in H$, so $g_i H = g_j H$, so $i = j$. Note that $|Hg_i| = |H|$, so $\{Hg_i : i \in [n]\}$ is a partition

of G .

Let $g \in G$, so there exists $h \in H$ and $i \in [n]$ such that $g = hg_i$, so $\mathbb{C}[G]$ is spanned by $\{g_i\}$. Suppose $\sum_{i \in [n]} x_i g_i = 0$ for $x_i \in \mathbb{C}[H]$. Enumerate $H = \{h_j : j \in [m]\}$ and write $x_i = \sum_j \lambda_{ij} h_j$, so $\sum_{i \in [n]} \sum_{j \in [m]} \lambda_{ij} h_j g_i = 0$. Since the $\{h_j g_i : i \in [n], j \in [m]\}$ is an indexing of G (which is a basis of $\mathbb{C}[G]$), we have that $\lambda_{ij} = 0$ for all i, j , from which it follows that the g_i form a basis.

2 Composition Series and Jordan Holder

Claim: Let G be finite. Then G has a composition series.

Proof: If $G = 1$, we're done. Else, pick maximal $H \triangleleft G$ (which exists since G is finite and $1 \triangleleft G$). I claim that G/H is simple. Indeed, if $1 < K \triangleleft G/H$, then $H < K' \triangleleft G$, where K' is the complete preimage of K , absurd. By IH, let

$$1 = H_0 \triangleleft H_1 \triangleleft H_2 \triangleleft \dots \triangleleft H_r = H$$

be a composition series of H . Then adding G to the end of the series gives a composition series for G .

Claim: The composition factors of G are the same length and unique up to permutation.

Proof: Suppose

$$\begin{aligned} 1 &= G_0 \triangleleft G_1 \triangleleft G_2 = G \\ 1 &= G'_0 \triangleleft G'_1 \triangleleft \dots \triangleleft G'_s = G \end{aligned}$$

for some $s \geq 2$ are composition series of G . Then we have the series

$$1 = G'_0 \cap G_1 \trianglelefteq G'_1 \cap G_1 \trianglelefteq \dots \trianglelefteq G'_s \cap G_1 = G_1.$$

Since G_1 is simple, there exists $t \in [s-1]$ such that $G'_i \cap G_1 = 1$ for $0 \leq i \leq t$ and $G'_i \cap G_1 = G_1$ for $i > t$. Hence, $G_1 \leq G'_i$ for $i > t$, so we have

$$G'_{t+1}/G_1 \triangleleft G'_{t+2}/G_1 \triangleleft \dots \triangleleft G'_s/G_1 = G_2/G_1,$$

so $s - t \leq 2$ (since G_2/G_1 is simple). Consider the series

$$1 = G'_0 G_1 / G_1 \trianglelefteq G'_1 G_1 / G_1 \trianglelefteq \dots \trianglelefteq G'_s G_1 / G_1 = G_2 / G_1.$$

Note $G'_i G_1 / G_1 \cong G'_i / (G'_i \cap G_1) \cong G'_i$ for $i \leq t$, so $t \leq 1$. If $t = 0$, we have $s \leq 2$, so $s = 2$. If $t = 1$, then $G'_1 G_1 = G_2$ and $G'_1 \cong G_2 / G_1$.

Claim: Let $H \leq G$ and $K \trianglelefteq G$. Then $HK/K \cong H/(H \cap K)$.

Proof: Consider the natural map $H \rightarrow HK/K$, which is surjective. The kernel is $H \cap K$, so $H \cap K \trianglelefteq H$ and the isomorphism follows.

3 Galois Theory

Claim: Let finite G act transitively on finite X . Then $|X| \mid |G|$.

Proof: Fix $x_0 \in X$. We have $|G_{x_0}| \cdot |Gx_0| = |G|$. Since $|Gx_0| = |X|$, we have $|G|/|X| = |G_{x_0}| \in \mathbb{Z}$, as desired.

Claim: Any 5 cycle and 2 cycle together generate S_5 .

Proof: Let $\sigma = (12345)$ be a 5 cycle and $\tau = (12), (13)$. Other cases need not be considered since they are equivalent up to relabeling-possibilities of τ are from casing on the relative positions of the numbers being swapped by τ in σ .

Note that $(12345)^2 = (13524)$, so the second case is the same as the first. It is known that a $\langle (12345), (12) \rangle = S_5$.

Claim: Let E/F be Galois and suppose $E/K/F$. Then the following are equivalent:

- (i). K/F is Galois.
- (ii). $\sigma(K) = K$ for all $\sigma \in \text{Aut}(E/F)$.
- (iii). $\text{Aut}(E/K) \trianglelefteq \text{Aut}(E/F)$ and $\text{Aut}(E/F)/\text{Aut}(E/K) \cong \text{Aut}(K/F)$.

Proof: (i) $\rightarrow$ (ii). Since K/F is Galois, K is the splitting field of some separable $f \in F[x]$. Let f have roots α_i for $i \in [n]$. $\sigma(\alpha_i) = \alpha_j$ for some $j \in [n]$, so $\sigma(K) \leq K$. Similarly, $\sigma^{-1}(K) \leq K$, so $\sigma(K) = K$.

(ii) $\rightarrow$ (iii). Define $\varphi : \text{Aut}(E/F) \rightarrow \text{Aut}(K/F)$ by $\varphi(\sigma) = \sigma|_K$. The homomorphism φ is well-defined by (ii). Since E/F is Galois, let E be the splitting field of some separable $g \in F[x]$. Treat $g \in K[x]$, so given $\sigma' \in \text{Aut}(K/F)$, we can extend to an isomorphism $\sigma \in \text{Aut}(E/F)$. Hence, φ is surjective. $\ker \varphi$ is precisely $\text{Aut}(E/K)$, so $\text{Aut}(E/K) \trianglelefteq \text{Aut}(E/F)$ and $\text{Aut}(E/F)/\text{Aut}(E/K) \cong \text{Aut}(K/F)$.

(iii) $\rightarrow$ (i). Taking orders of both sides, $|\text{Aut}(K/F)| = |\text{Aut}(E/F)|/|\text{Aut}(E/K)| = [E : F]/[E : K] = [K : F]$.

Claim: Let $F = K_0 \leq K_1 \leq K_2 \leq \dots \leq K_s = K$ be such that $K_{i+1} = K_i(\sqrt[n_i]{a_i})$ for some $n_i \in \mathbb{Z}^+$ and $a_i \in K_i$. I claim that K is contained in an extension

L/F such that there exists a chain $F = L_0 \leq L_1 \leq L_2 \leq \dots \leq L_t = L$, where L_{i+1}/L_i is cyclic.

Proof: Let L be the Galois closure of K . Given $\sigma \in \text{Aut}(L/F)$, consider the chain $F = \sigma K_0 \leq \sigma K_1 \leq \dots \leq \sigma K_s = \sigma K$, so $\sigma(K_{i+1}) = \sigma(K_i)(\sigma(\alpha_i))$. I claim that the composition of two radical extensions is a radical extension. Indeed, if $F = K_0 \leq K_1 \leq \dots \leq K_s = K$ and $F = K'_0 \leq K'_1 \leq \dots \leq K'_t = K'$ are radical extensions, then $F = K_0 \leq K_1 \leq \dots \leq K_s = K_s K'_0 \leq K_s K'_1 \leq \dots \leq K_s K'_t$, where $K_s K'_{j+1}/K_s K'_j$ is a simple radical extension.

Note that $\text{Aut}(L/F)$ is finite, so the composition $L = \bigvee_{\sigma \in \text{Aut}(L/F)} \sigma(K)$ is a radical extension. Since we are working over characteristic 0, let F' be a Cyclotomic extension of F containing the n_i th roots of unity. Then $F \leq F' L_0 \leq F' L_1 \leq \dots \leq F' L_m = F' L$. Since $F', L/F$ is Galois, so is $F' L/F$. Hence, L_{i+1}/L_i is cyclic.

Claim: Let $f \in F[x]$ be solvable by radicals. Let E/F be the splitting field of f , so E is a radical extension. Let L/E be such that L is Galois and there exists a chain $F = L_0 \leq L_1 \leq \dots \leq L_s = L$, where L_{i+1}/L_i is cyclic. Let $G_i = \text{Aut}(L/L_i)$. Then $\text{Aut}(L_{i+1}/L_i) \cong G_i/G_{i+1}$, so $G_0 = \text{Aut}(L/F)$ is solvable. Then $\text{Aut}(E/F)$ as a quotient of $\text{Aut}(L/F)$ is solvable.

Claim: Let E/F be Galois and $E/K_1/K_2/F$ such that K_1/K_2 is Galois. Then $\text{Aut}(K_1/K_2) \cong \text{Aut}(E/K_2)/\text{Aut}(E/K_1)$.

Proof: We have $E/K_1/K_2$ such that $E/K_1, K_1/K_2, E/K_2$ are Galois. Then $\text{Aut}(K_1/K_2) \cong \text{Aut}(E/K_2)/\text{Aut}(E/K_1)$, as desired.

4 Semidirect Product

Claim: Let $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. Let $H \rtimes K = \{(h, k) : h \in H, k \in K\}$ have the product given by $(h_1, k_1) \cdot (h_2, k_2) = (h_1 \varphi(k_1)(h_2), k_1 k_2)$. I claim that this product gives $H \rtimes K$ a group structure.

Proof: We have $(1, 1)(h, k) = (h, k) = (h, k)(1, 1)$, so $(1, 1)$ is the identity. Further, $(h, k)(\varphi(k^{-1})(h^{-1}), k^{-1}) = (1, 1) = (\varphi(k^{-1})(h^{-1}), k^{-1})(h, k)$, so inverses exist. Finally,

$$\begin{aligned} ((h_1, k_1) \cdot (h_2, k_2)) \cdot (h_3, k_3) &= (h_1 \varphi(k_1)(h_2), k_1 k_2) \cdot (h_3, k_3) \\ &= (h_1 \varphi(k_1)(h_2) \varphi(k_1 k_2)(h_3), k_1 k_2 k_3) = (h_1 \varphi(k_1)(h_2) \varphi(k_1)(\varphi(k_2)(h_3)), k_1 k_2 k_3) \\ &= (h_1 \varphi(k_1)(h_2 \varphi(k_2)(h_3)), k_1 k_2 k_3) = (h_1, k_1) \cdot (h_2 \varphi(k_2)(h_3), k_2 k_3) \\ &= (h_1, k_1) \cdot ((h_2, k_2) \cdot (h_3, k_3)), \end{aligned}$$

so the product is associative.

I claim that $H = \{(h, 1) : h \in H\} \trianglelefteq H \rtimes K$. Indeed, $(h_1, 1) \cdot (h_2, 1) = (h_1 h_2, 1)$ and the map $h \mapsto (h, 1)$ gives a group isomorphism. We have $(h', k') \cdot (h, 1) \cdot (\varphi(k'^{-1})(h'^{-1}), k^{-1}) = (h' \varphi(k')(h), k')(\varphi(k'^{-1})(h'^{-1}), k^{-1}) = (h' \varphi(k')(h) \varphi(k') \varphi(k'^{-1})(h'^{-1}), 1) = (h' \varphi(k')(h) h'^{-1}, 1)$. In particular, if $h' = 1$, $(1, k')$ acts as the action of K on H .

I claim $(h, k) = (h, 1) \cdot (1, k)$ (easily verifiable). Then conjugating by (h, k) first acts by k , then conjugates by h . I claim $K = \{(1, k) : k \in K\}$ is an identification of K . Indeed, $(1, k_1) \cdot (1, k_2) = (1, k_1 k_2)$ and the map $k \mapsto (1, k)$ is bijective.

We have $H \cap K = 1$ and $HK = H \rtimes K$.

Conversely, suppose for a group G that $H \trianglelefteq G$, $HK = G$, and $H \cap K = 1$. I claim that $G \cong H \rtimes K$ for some $\varphi : K \rightarrow \text{Aut}(H)$.

Indeed, we take $\varphi(k)(h) = khk^{-1}$, which is well-defined since $H \trianglelefteq G$.